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D15C/58

Experiment no. 9

Aim

To implement Recurrent Neural Network (RNN) using LSTM for classification on Breast Cancer dataset.

Theory

RNN is a deep learning model used for sequential data. It maintains memory of previous inputs using loops.

LSTM (Long Short-Term Memory) is an advanced type of RNN that solves the problem of vanishing gradients and can remember long-term dependencies.

It uses gates:

- Forget Gate
- Input Gate
- Output Gate

In this experiment, tabular data is reshaped into sequence format to apply LSTM.

Code

With hyperparameter tuning

```
def build_lstm(hp):
    model = models.Sequential()

    model.add(layers.LSTM(
        units=hp.Int('units', 32, 128, step=32),
        input_shape=(30,1)
    ))

    model.add(layers.Dense(1,
activation='sigmoid'))

    model.compile(optimizer='adam',
loss='binary_crossentropy', metrics=['accuracy'])

    return model

tuner = RandomSearch(build_lstm,
objective='val_accuracy', max_trials=3)

tuner.search(X_train_rnn, y_train, epochs=5,
validation_data=(X_test_rnn, y_test))
```

Output:

```
INFO Reloading Tuner from ./untitled_project/tuner0.json
```

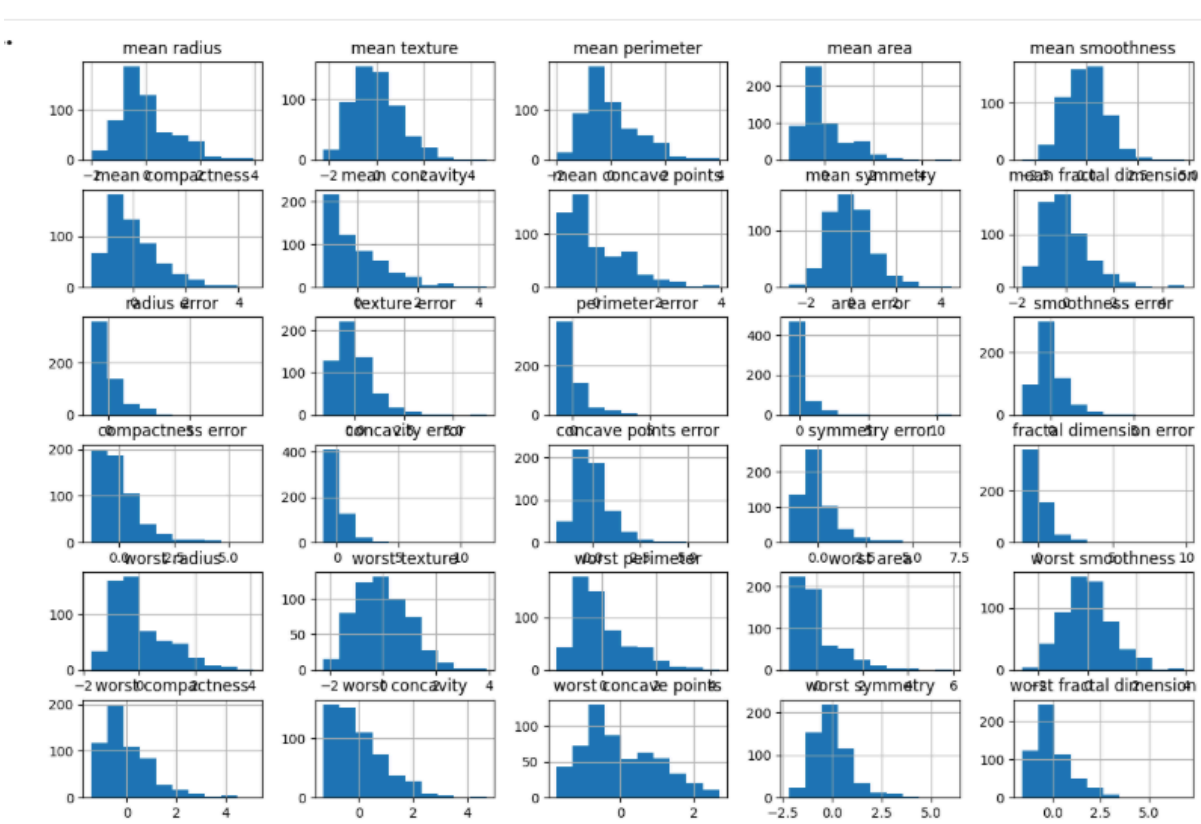
Without hyperparameter tunning

```
model = models.Sequential([
    layers.LSTM(32, input_shape=(30,1)),
    layers.Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam',
loss='binary_crossentropy', metrics=['accuracy'])

model.fit(X_train_rnn, y_train, epochs=10,
validation_data=(X_test_rnn, y_test))
```

Output:

```python
model.fit(X_train_rnn, y_train, epochs=10, validation_data=(X_test_rnn, y_test))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not p
  super().__init__(**kwargs)
Epoch 1/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 6s 93ms/step - accuracy: 0.9121 - loss: 0.6095 - val_accuracy: 0.88
Epoch 2/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 2s 35ms/step - accuracy: 0.8967 - loss: 0.4619 - val_accuracy: 0.91
Epoch 3/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 51ms/step - accuracy: 0.8857 - loss: 0.3116 - val_accuracy: 0.87
Epoch 4/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 46ms/step - accuracy: 0.9143 - loss: 0.2632 - val_accuracy: 0.86
Epoch 5/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 53ms/step - accuracy: 0.9055 - loss: 0.2582 - val_accuracy: 0.88
Epoch 6/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 35ms/step - accuracy: 0.9077 - loss: 0.2552 - val_accuracy: 0.87
Epoch 7/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 48ms/step - accuracy: 0.9077 - loss: 0.2509 - val_accuracy: 0.89
Epoch 8/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 70ms/step - accuracy: 0.9143 - loss: 0.2428 - val_accuracy: 0.89
Epoch 9/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 56ms/step - accuracy: 0.9143 - loss: 0.2359 - val_accuracy: 0.90
Epoch 10/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 1s 76ms/step - accuracy: 0.9187 - loss: 0.2225 - val_accuracy: 0.91
<keras.src.callbacks.history.History at 0x7a8cd9e91610>
```



Conclusion

RNN and LSTM are powerful models for sequential data. When adapted properly, they can also be used for structured datasets.